PlutoPack V2 — sourced BOM & cost

Real JLCPCB/LCSC part numbers (Basic preferred) with Digikey/other for parts JLC lacks or large/mechanical parts. Prices read off LCSC/JLCPCB product pages on **2026-07-04**, qty-1 unit pricing unless noted. 108 placements / 46 line items. **Nothing here changes the KiCad project.**

Component subtotal (qty-1): $33.15

Semiconductors $16.39 · Passives $5.07 · Connectors/switches/mech $11.69

+ Display panel ~$6–8 (separate purchase) → **~$40 all-in parts**.

Excludes: PCB fabrication, JLC PCBA assembly/setup, and JLC per-unique-**Extended**-part handling fees (~$3 each; there are ~18 Extended line items → order Basic-only where you can, or hand-place Extended parts). Prices are indicative — **verify live stock & price at order** (esp. C183096 protector: low stock).

Semiconductors

Ref(s)	Qty	Value / MPN	Footprint	JLC LCSC# / Digikey PN	Class	Unit $	Ext $	Notes
U1	1	BQ24075RGTR — Li-ion charger	QFN-16-EP 3×3	JLC C15464	Ext	0.8984	0.898	
U4	1	BQ27441DRZR-G1A — fuel gauge	SON-12-EP 2.5×4 ⚠	JLC C473397	Ext	3.1234	3.123	fp corr: not 2.5×2.5
U3,U5	2	TPS63020DSJR — buck-boost	VSON-14-EP 3×4 ⚠	JLC C15483	Ext	0.8533	1.707	fp corr: not 2.5×2.5
U2	1	TSC2007IPWR — touch ctrl	TSSOP-16	JLC C38398	Ext	1.3122	1.312	
U7	1	TXS0108EPWR — level shifter	TSSOP-20	JLC C17206	Ext	0.3864	0.386	
U6	1	BQ29700DSER — 1-cell protector	WSON-6 1.5×1.5	JLC C183096	Ext	0.5748	0.575	low JLC stock — verify
Q1	1	AO3401A — P-MOSFET	SOT-23	JLC C15127	Basic	0.0989	0.099	
Q2,Q3	2	AO3400A — N-MOSFET	SOT-23	JLC C20917	Basic	0.0838	0.168	
D1-D4	4	KT-0603R — red LED	LED_0603	JLC C2286	Basic	0.0073	0.029	
D5	1	B5819W — Schottky	SOD-123 (assign)	JLC C8598	Basic	0.0302	0.030	fp assign
A2	1	Raspberry Pi Pico module	Pico SMD module	JLC C7203002	Ext	8.0661	8.066	or hand-solder/Digikey
Semiconductor subtotal							**16.39**	

Passives

Ref(s)	Qty	Value / MPN	Footprint	JLC LCSC# / Digikey PN	Class	Unit $	Ext $	Notes
R7,R10,R13,R19-R31	16	10 kΩ 1%	0603	JLC C25804	Basic	0.0020	0.032	
R34-R40	7	0 Ω jumper	0603	JLC C21189	Basic	0.0028	0.020	
R1-R4,R32,R33	6	10 Ω 1% (backlight)	0603	JLC C22859	Basic	0.0025	0.015	
R11,R12,R14,R15,R43,R44	6	1 kΩ 1%	0603	JLC C21190	Basic	0.0027	0.016	
R17,R18,R49,R50	4	5.1 kΩ 1%	0603	JLC C23186	Basic	0.0028	0.011	
R6,R9	2	180 kΩ 1%	0603	JLC C22827	Ext	0.0028	0.006	
R41,R42	2	100 kΩ 1%	0603	JLC C25803	Basic	0.0025	0.005	

R5	1	1 MΩ 1%	0603	JLC C22935	Basic	0.0027	0.003	see crit-3: should be 1.62 M
R8	1	1.6 MΩ 1%	0603	VERIFY (0603WAF1604)	Ext	0.0050	0.005	see crit-3: should be 1.0 M
R48	1	2.2 kΩ 1%	0603	JLC C4190	Basic	0.0024	0.002	
R45	1	330 Ω 1%	0603	JLC C23138	Basic	0.0074	0.007	
R46,R47	2	5 MΩ → 5.1 MΩ (sub)	0603	JLC C13320	Ext	0.0024	0.005	5 M not std; 5.1 M sub
R16	1	10 mΩ sense 1%	2512 3W ⚠	JLC C2994644	Ext	0.0601	0.060	fp corr: 0603→2512
C1,C2,C9,C12,C15,C20	6	100 nF / 0.1 µF X7R 50V	0603	JLC C14663	Basic	0.0234	0.140	100n & 0.1u same part
C19	1	0.47 µF X7R 25V	0603	JLC C1623	Basic	0.0088	0.009	
C7,C18	2	1 µF X5R 50V	0603	JLC C15849	Basic	0.1295	0.259	
C17	1	4.7 µF X5R 16V	0603	JLC C19666	Basic	0.0997	0.100	
C3,C4,C10,C11	4	10 µF X5R 10V	0603	JLC C19702	Basic	0.0550	0.220	10V — derate DC bias
C5,C16	2	100 µF tantalum 10V	EIA-7343 (D)	JLC C110549	Ext	0.6320	1.264	
L1,L2	2	1.2 µH → 1.5 µH (sub)	12.5×13.5 mm	JLC C2047209	Ext	1.4300	2.860	SRP1265A-1R5M, Isat 45A
FB1,FB2	2	Ferrite 120Ω@100MHz 2A	0603	JLC C14709	Basic	0.0145	0.029	value was unset
C6,C8,C13,C14	4	VALUE NOT SET	0603 (assign)	None	—	—	—	assign value (e.g. 10 µF) + fp
Passives subtotal							**5.07**	

Connectors · switches · mechanical

Ref(s)	Qty	Value / MPN	Footprint	JLC LCSC# / Digikey PN	Class	Unit $	Ext $	Notes
J1	1	2×10 socket 2.54 mm	PinSocket 2x10	JLC C30867	Ext?	0.1160	0.116	class verify
J4	1	1×6 header 2.54 mm	PinSocket 1x06	JLC C37208	Ext?	0.0150	0.015	class verify
J2	1	FFC 50P 0.5 mm horiz, bottom	FFC-50 0.5mm	JLC C9355	Ext	0.5360	0.536	~Amphenol F32Q equiv
J5	1	USB-C receptacle 6P power	USB-C 6P (assign)	JLC C456012	Ext	0.0380	0.038	fp assign
J3	1	18650 holder (Keystone 1043)	THru-hole holder	Digikey 36-1043-ND	—	2.9900	2.990	fp corr: 1043 TH (not 1042 SMT)
SW1,SW3-SW8	7	Tact switch C&K KSC621GLFS	6.2mm SMD (KSC6)	JLC C221759	Ext	0.7900	5.530	budget: TS-1187A C318884 $0.011 (re-fp)
SW2	1	Rotary encoder Alps EC12E24244	EC12E vert 20mm	Mouser EC12E2424407	—	1.6600	1.660	price verify (Mouser)
SW10	1	Slide SPDT C&K OS102011MS2Q	SPDT slide TH	Digikey CKN9565-ND	—	0.8000	0.800	
H1-H4	4	M3 mounting hole	plated hole	— (no part)	—	—	—	
Connectors/switches subtotal							**11.69**	

Separate purchase (not board-assembled)

Ref(s)	Qty	Value / MPN	Footprint	JLC LCSC# / Digikey PN	Class	Unit $	Ext $	Notes
Display	1	CH320QV15A-T bare 2.4" 240×320 ILI9341 + RTP	mates to J2 FFC	AliExpress / Buydisplay	—	—	—	not a JLC-assembled part; confirm FFC pinout

Footprint corrections & assignments needed

- **Corrections:** BQ27441 → **SON-12-EP 2.5×4 mm** (DRZ; the 2.5×2.5 is the WCSP part). TPS63020**DSJR** → **VSON-14-EP 3×4 mm** (verify your custom QFN14 land). R16 10 mΩ → **2512** (0603 impractical for a shunt). 18650 holder → Keystone **1043** (through-hole), not 1042 (SMT).
- **Assign (currently blank in schematic):** D5 → SOD-123; J5 → USB-C 6P land; J3 → 18650 holder; and R42, R45, R46, R47, R48, R49, R50, C7 → 0603.
- **C6, C8, C13, C14:** value is unset — assign a value (e.g. 10 µF) and 0603 footprint, then source (not costed above).

Key substitutions & cautions

- 10 mΩ sense: 2512 3 W metal-element shunt (no Basic option).
- 1.2 µH inductor: no ~12 mm 1.2 µH at JLC → 1.5 µH SRP1265A-1R5M (the value TI's TPS63020 datasheet lists as typical; Isat 45 A). For a true 1.2 µH use Coilcraft XAL1010-122 via Digikey.
- 5 MΩ: not a standard value → 5.1 MΩ substituted. 1.6 MΩ: LCSC C# unconfirmed — verify. (Both R5/R8 values should change anyway per review crit-3.)
- Tact switches: C&K KSC621GLFS matches the KSC6 6.2 mm land ($0.79×7). Budget TS-1187A (C318884, ~$0.011) needs a different footprint → re-foot if cost matters (saves ~$5.4).
- Encoder: Alps EC12E2424407 is Mouser/Newark stocked (not Digikey); price to verify.
- Display: Adafruit sells only the breakout (#2478, ~$30); the bare CH320QV15A-T FFC panel is AliExpress/Buydisplay (~$6–8). Confirm the FFC pinout matches J2.
- Basic parts: all generic R/C, the two MOSFETs, LED, Schottky, and ferrite are JLC Basic. The six TI ICs, Pico, tantalum, inductor, 180 k, 5.1 M, tantalum, and the connectors are Extended.